

Foundations for programming languages 2: Answers for exercises

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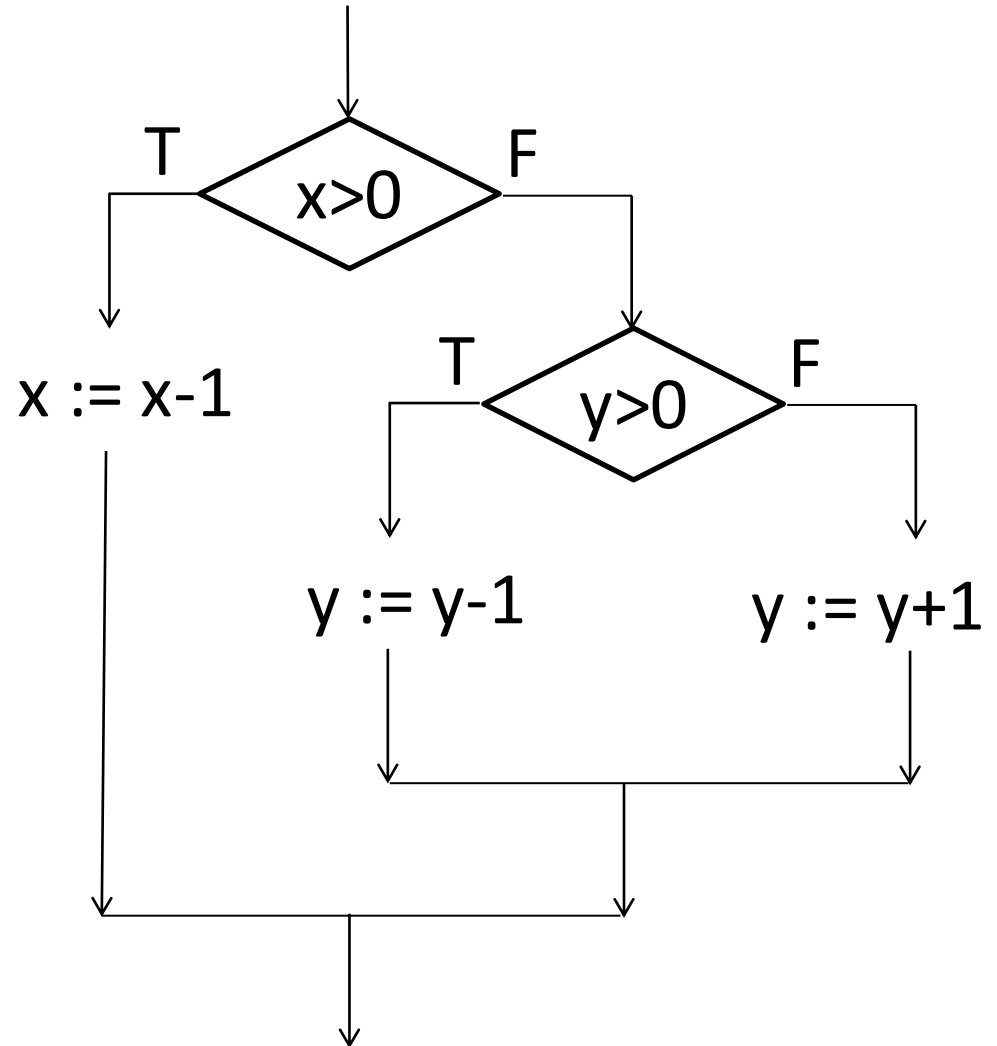
Ex. 1

Illustrate the control flow graph of the following program fragment.

```
if  $x > 0$  then  $x := x - 1$   
else if  $y > 0$  then  $y := y - 1$   
      else  $y := y + 1$ 
```

An answer

if $x > 0$ **then** $x := x - 1$
else if $y > 0$ **then**
 $y := y - 1$
else $y := y + 1$



Ex. 2

Illustrate the control flow graph of the following program fragment.

$y := 3;$

case x of

1 : $y := 1;$

2 : $y := x * 2;$

3 : **if** $z = 0$ **then**

$y := y * y$

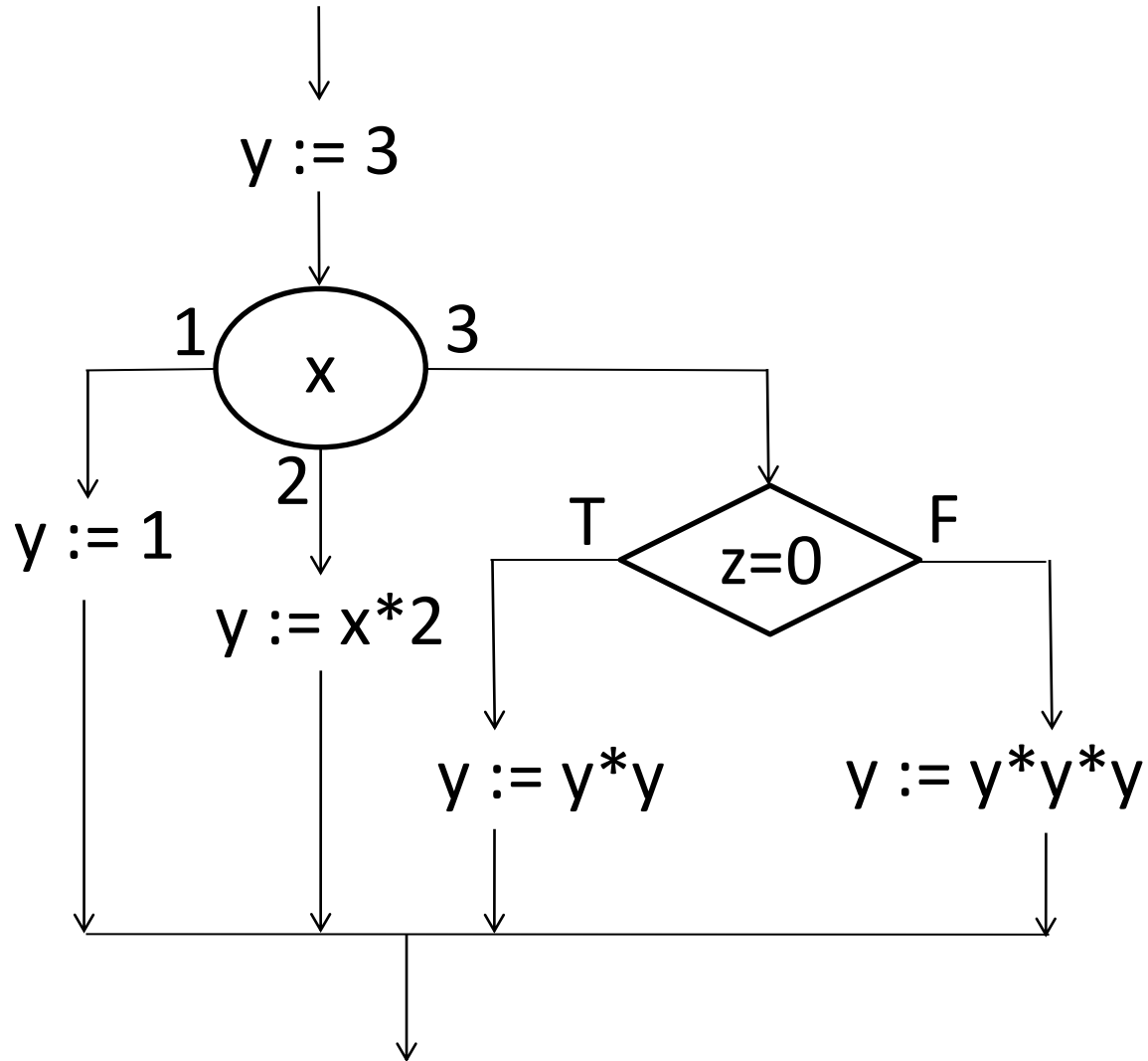
else

$y := y * y * y$

end

An answer

```
y := 3;  
case x of  
  1 : y := 1;  
  2 : y := x * 2;  
  3 : if z = 0 then  
        y := y * y  
      else  
        y := y * y * y  
      end  
end
```



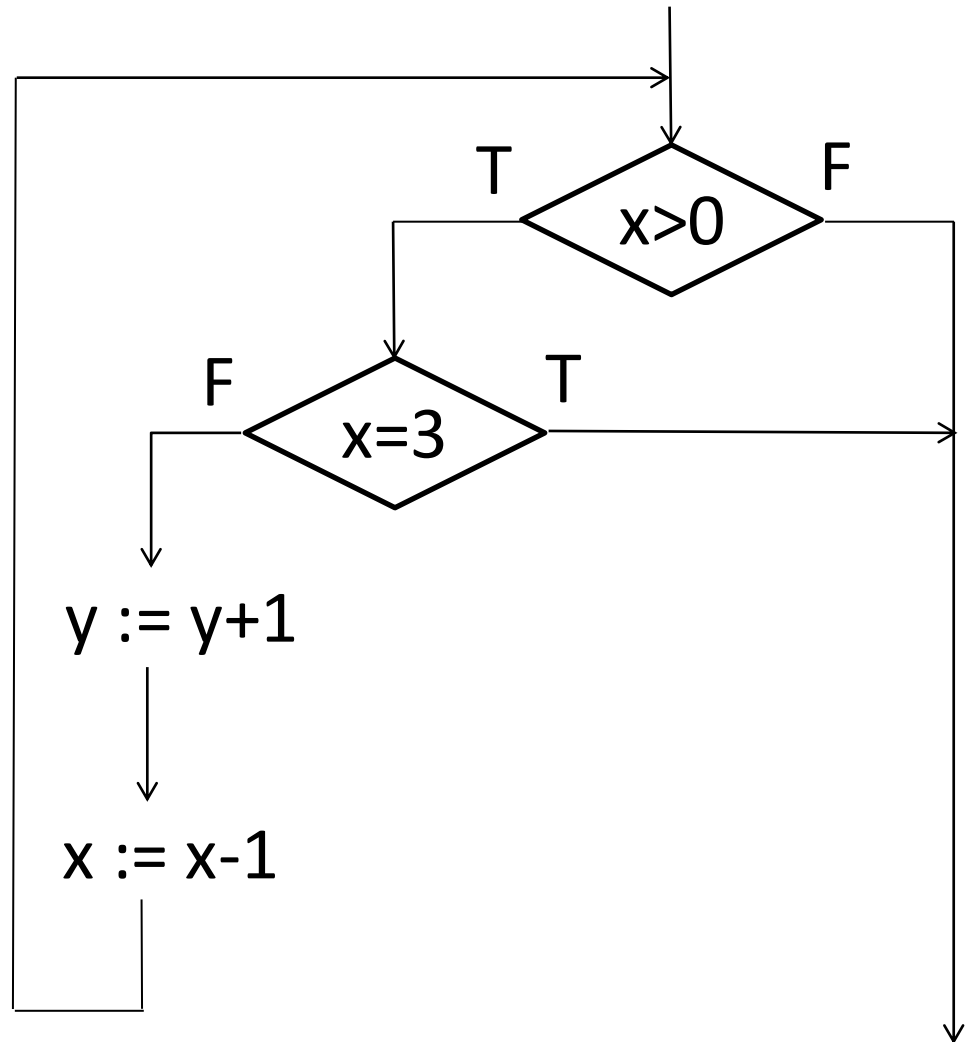
Ex. 3

Illustrate the control flow graph of the following program fragment.

```
while x>0 do  
  begin  
    if x=3 then  
      break;  
    y := y + 1;  
    x := x - 1  
  end
```

An answer

```
while  $x > 0$  do  
  begin  
    if  $x = 3$  then  
      break;  
     $y := y + 1$ ;  
     $x := x - 1$   
  end
```



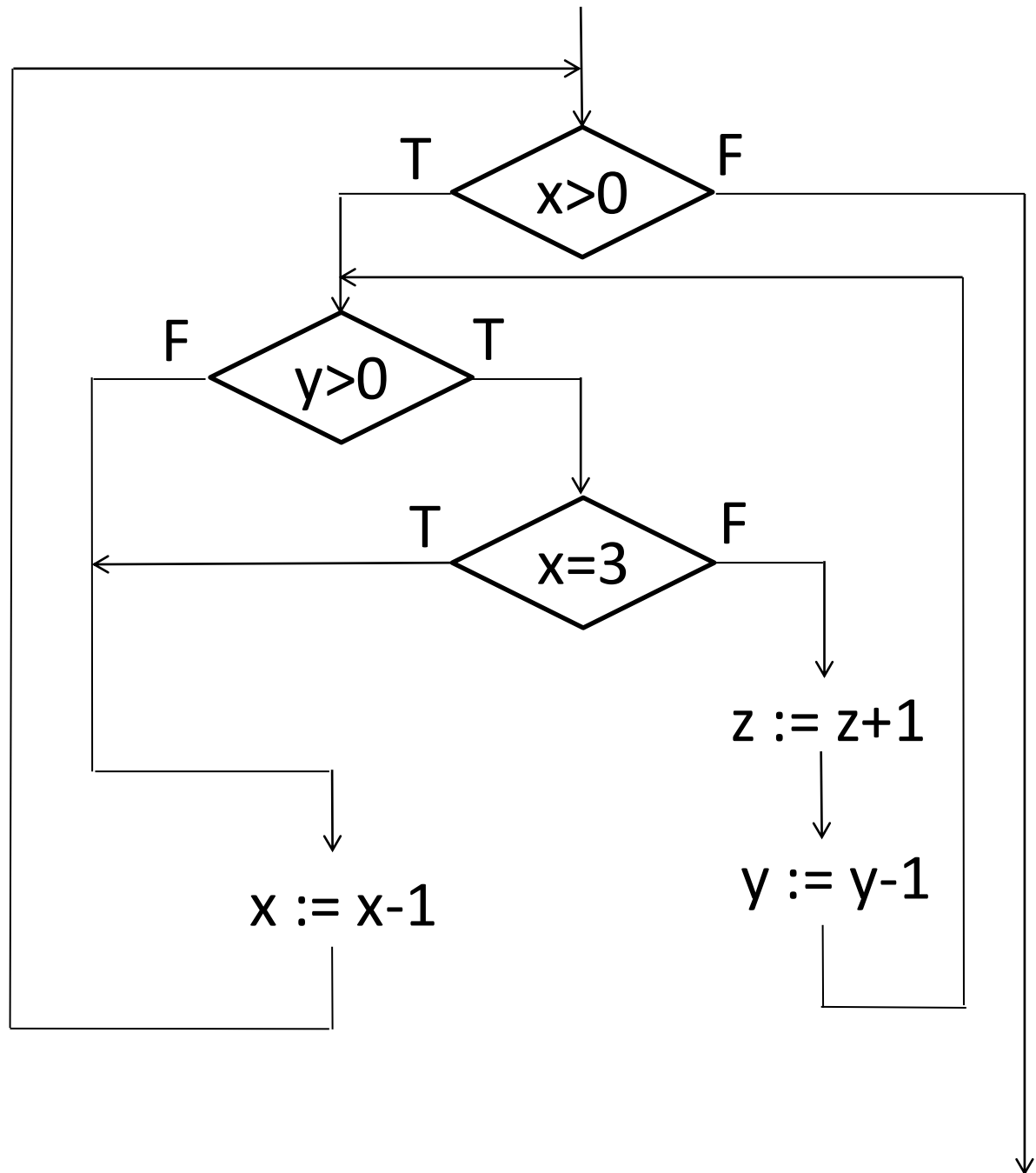
Ex. 4

Illustrate the control flow graph of the following program fragment.

```
while x>0 do  
  begin  
    while y>0 do  
      begin  
        if x=3 then  
          break;  
        z := z + 1;  
        y := y - 1  
      end;  
    x := x - 1  
  end
```

An answer

```
while x>0 do
  begin
    while y>0 do
      begin
        if x=3 then
          break;
        z := z + 1;
        y := y - 1
      end;
      x := x - 1
    end
  end
```

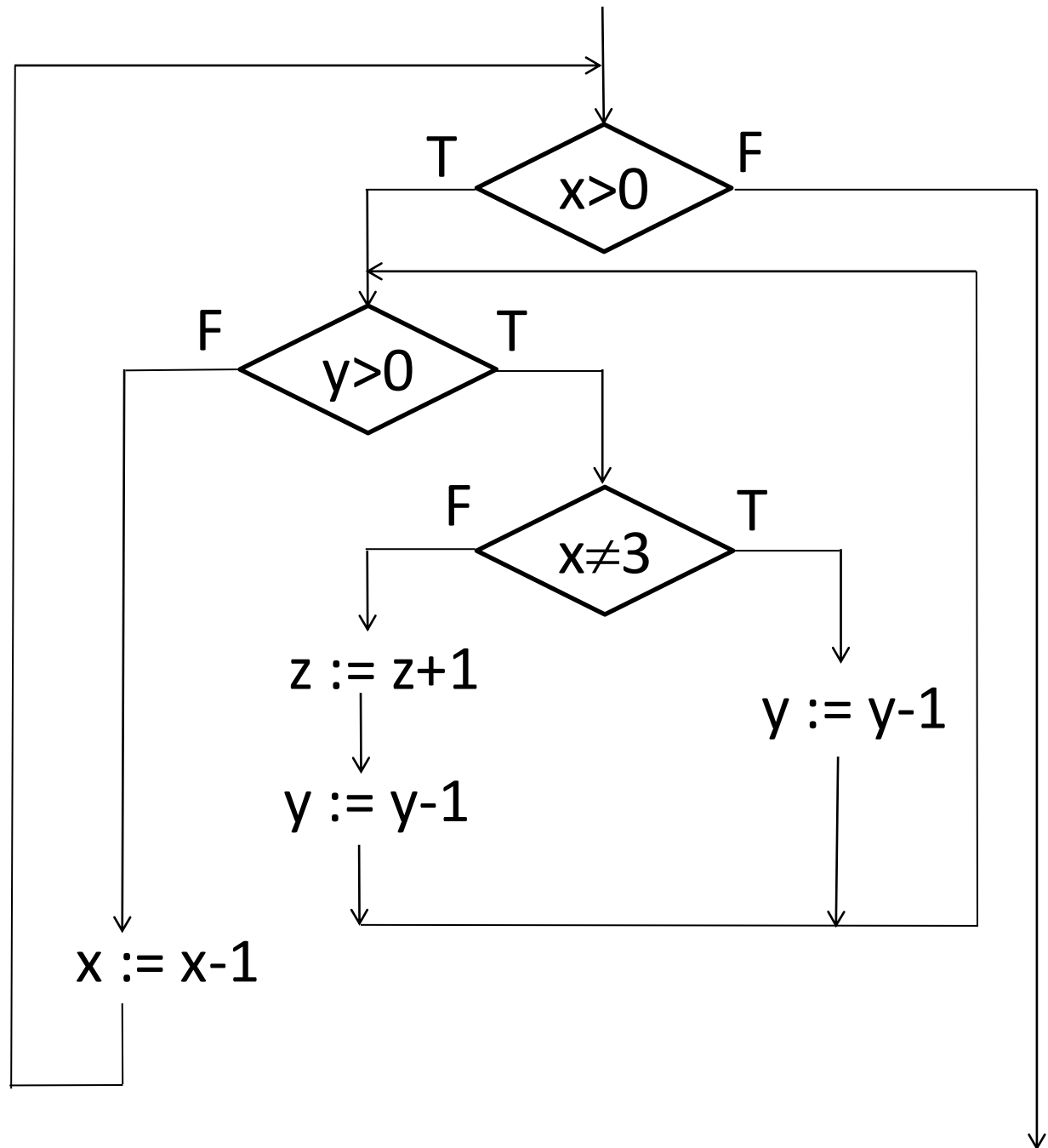


Ex. 5 Illustrate the control flow graph of the following program fragment.

```
while  $x > 0$  do  
  begin  
    while  $y > 0$  do  
      begin  
        if  $x \neq 3$  then  
          begin  
             $y := y - 1;$   
            continue  
          end  
           $z := z + 1; y := y - 1$   
        end;  
         $x := x - 1$   
      end  
    end
```

An answer

```
while x>0 do
  begin
    while y>0 do
      begin
        if x≠3 then
          begin
            y := y - 1;
            continue
          end
        z := z + 1;
        y := y - 1
      end;
      x := x - 1
    end
  end
```



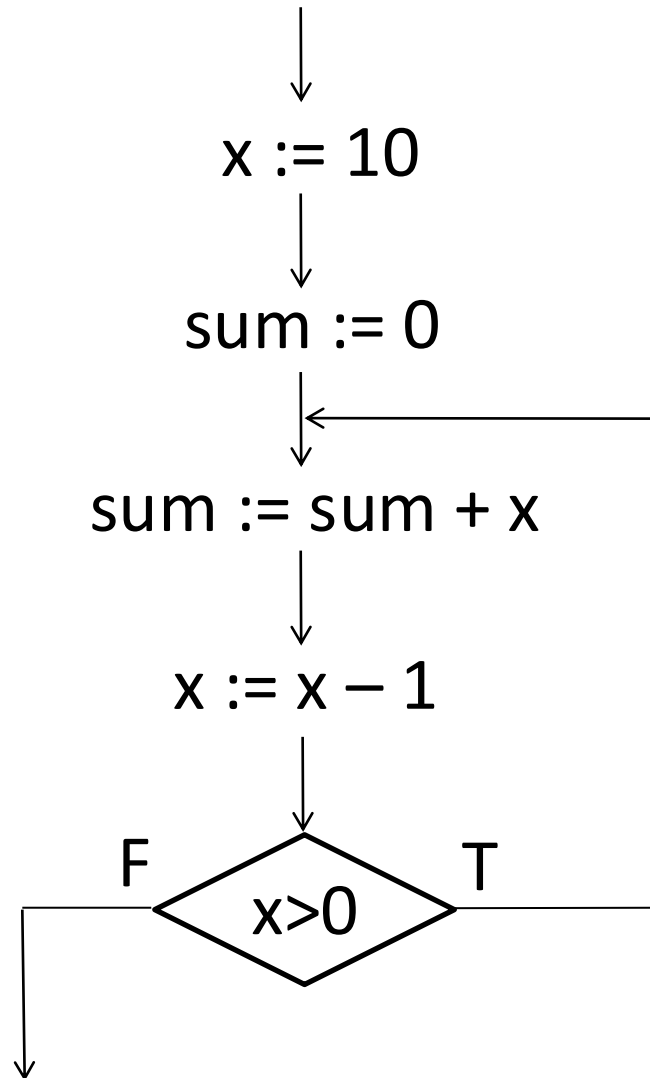
Ex. 6

Illustrate the control flow graph of the following program fragment.

```
x := 10;  
sum := 0;  
L: sum := sum + x;  
x := x - 1;  
if x > 0 then  
    goto L
```

An answer

```
x := 10;  
sum := 0;  
L: sum := sum + x;  
x := x - 1;  
if x > 0 then  
    goto L
```



Exercise 7

How many entries and exits do the two **if** statements have?

```
while  $x > 0$  do  
  begin  
    while  $y > 0$  do  
      begin  
        if  $x = 3$  then  
          break;  
        L:  $z := z + 1$ ;  
         $y := y - 1$   
      end;  
       $x := x - 1$ ;  
      if  $x = 2$  then goto L  
    end
```

An answer

```
while x>0 do
  begin
    while y>0 do
      begin
        if x=3 then
          break;
        L: z := z + 1;
        y := y - 1
      end;
      x := x - 1;
      if x = 2 then goto L
    end
  end
```

